

Anti-cancer properties of phenolics from apple waste on colon carcinogenesis in vitro.



Colorectal cancer is one of the most common cancers in Western countries. The World Health Organisation identifies diet as a critical risk factor in the development and progression of this disease and the protective role of high levels of fruit and vegetable consumption. Several studies have shown that apples contain several phenolic compounds that are potent anti-oxidants in humans. However, little is known about other beneficial properties of apple phenolics in cancer. We have used the HT29, HT115 and CaCo-2 cell lines as in vitro models to examine the effect of apple phenolics (0.01-0.1% apple extract) on key stages of colorectal carcinogenesis, namely; DNA damage (Comet assay), colonic barrier function (TER assay), cell cycle progression (DNA content assay) and invasion (Matrigel assay). Our results indicate that a crude extract of apple phenolics can protect against DNA damage, improve barrier function and inhibit invasion ($p < 0.05$). The anti-invasive effects of the extract were enhanced with twenty-four hour pretreatment of cells ($p < 0.05$). We have shown that a crude apple extract from waste, rich in phenolic compounds, beneficially influences key stages of carcinogenesis in colon cells in vitro.